

In this paper, all problem in the Silverman will be solved in complete form.

#7, 1. a,

Let $C: z(t) = e^{it}$ ($-\pi \leq t \leq \pi$). Evaluate

$$\int_C z^2 dz = \int_{-\pi}^{\pi} e^{2it} \cdot i \cdot e^{it} dt = i \int_{-\pi}^{\pi} e^{3it} dt = i \cdot \left[\frac{1}{3} i \cdot e^{3it} \right]_{-\pi}^{\pi} = 0$$

$$\int_C \frac{1}{z} dz = \int_{-\pi}^{\pi} \frac{1}{e^{it}} \cdot i \cdot e^{it} dt = 2\pi i.$$

$$\int_C \frac{1}{z^2} dz = \int_{-\pi}^{\pi} i \cdot \frac{1}{e^{2it}} dt = i \cdot \left[+i \cdot e^{-it} \right]_{-\pi}^{\pi} = -1 \cdot 0 = 0.$$

$$\int_C \overline{z} dz = \int_{-\pi}^{\pi} [\cos t - i \sin t] \cdot i \cdot e^{it} dt = i \int_{-\pi}^{\pi} \cos^2 t + \sin^2 t dt = 2\pi i.$$

#8.1.2. a)

Let $P(z)$ be a polynomial of degree n , then

$$\int_{|z|=2} \frac{P(z)}{(z-1)^{n+2}} dz = 0$$

Proof. By the Cauchy integral formula,

$$\int_{|z|=2} \frac{P(z)}{(z-1)^{n+2}} dz = \frac{P^{(n+1)}(1)}{(n+1)!} \cdot \frac{(n+1)!}{2\pi i} = 0.$$

#8.1.2. b)

Given $n, m \in \mathbb{N}$, show that $\int_C \frac{(n-1)! e^z}{(z-z_0)^n} dz = \int_C \frac{(m-1)! e^z}{(z-z_0)^m} dz$, where C is along z_0 .

Proof. By the Cauchy integral formula,

$$\int_C \frac{(n-1)! e^z}{(z-z_0)^n} dz = \frac{2\pi i}{(n-1)!} \cdot (n-1)! \cdot e^{z_0} = 2\pi i \cdot z_0.$$

Since the value is independent for n , so proved.

#8.1.3. Evaluate for $C: |z|=3$,

$$\int_C \frac{e^z}{z-2} dz = 2\pi i \cdot e^2.$$

$$\int_C \frac{e^z \sin z}{(z-2)^2} dz = 2\pi i \cdot (e^2 \sin 2 + e^2 \cdot \cos 2).$$

8.1.4.

$$\int_{|z|=2} \frac{1}{z^2-1} dz = \frac{1}{2} \int_{|z|=2} \frac{1}{z-1} - \frac{1}{z+1} dz = 0$$

$$\int_{|z|=3} \frac{z^2+3z-1}{(z-1)(z+2)} dz = \frac{1}{3} \int_{|z|=3} \frac{z^2+3z-1}{z-1} - \frac{z^2+3z-1}{z+2} dz = \frac{1}{3} \cdot 2\pi i \cdot (6) = 4\pi i$$

8.1.6.

observe that: for $|z|=1$, $\operatorname{Re} z = \frac{z+\bar{z}}{2} = \frac{1}{2} \cdot (z + \frac{1}{z})$.

$$\text{so } \int_{|z|=1} (\operatorname{Re} z)^2 dz = \frac{1}{4} \int_{|z|=1} \frac{(z^2+1)^2}{z^2} dz = \frac{1}{4} \cdot 2\pi i \cdot 8 = 4\pi i.$$

And, for $|z-1|=1$, $z=1+e^{i\theta}$ ($-\pi \leq \theta \leq \pi$), so $\bar{z}=1+e^{-i\theta}$, therefore

$$\bar{z} = 1 + \frac{1}{z-1} = \frac{z}{z-1}.$$

$$\text{Therefore } \int_{|z-1|=1} (\bar{z})^2 dz = \int_{|z-1|=1} \frac{z^2}{(z-1)^2} dz = 2\pi i \cdot 2 = 4\pi i.$$

And, for $|z-1|=1$, $\operatorname{Im} z = \operatorname{Im}(z-1) = \frac{1}{2i} \cdot (z - \bar{z}) = \frac{1}{2i} \cdot \left(\frac{z^2-2z}{z-1} \right)$, so

$$\int_{|z-1|=1} (\operatorname{Im} z)^2 dz = -\frac{1}{4} \cdot \int_{|z-1|=1} \frac{(z^2-2z)^2}{(z-1)^2} dz = -\frac{1}{4} \cdot 0 = 0.$$

#8.1.11. Evaluate the limits

$$\lim_{z \rightarrow 0} \frac{e^z - 1 - z}{z^2}$$

#8.1.12. If f is analytic at $z = z_0$, and $|f^{(n)}(z_0)| \leq n^k$ for each n .

Then f is an entire.

Proof. Since f is analytic at z_0 , all derivatives $f^{(n)}(z_0)$ exist and by the assumption,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Converses for any $z \in \mathbb{C}$, so f is analytic.

(More specifically,

$$\lim_{n \rightarrow \infty} \sup \frac{(n+1)^k}{(n+1)!} \cdot \frac{n!}{n^k} = \lim_{n \rightarrow \infty} \sup \left(1 + \frac{1}{n}\right)^k \cdot \frac{1}{n+1} = 0.$$

#8.1.13. Show that the following series are analytic.

$$f(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad (\operatorname{Re} z > 1), \quad f(z) = \sum_{n=1}^{\infty} \frac{1}{z^2 - n^2} \quad (z \neq \pm 1, \pm 2, \dots)$$

Proof. Recall that: $n^z = e^{z \cdot \ln n}$

$$2^x = e^{x \cdot \ln 2}$$

Section. 8.2.

#8.2.1. If $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is analytic in $|z| < R$, and $f(z)$ is real when $-R < z < R$. Show that all a_n are real, and $f(\bar{z}) = \overline{f(z)}$.

Proof. Let $g(z) = \overline{f(\bar{z})}$. So that

$$g(z) = \sum_{n=0}^{\infty} \overline{a_n} \cdot z^n$$

By the assumption, $f(x) - g(x) = \sum_{n=0}^{\infty} (a_n - \overline{a_n}) \cdot x^n = 0$ for all $x \in (-R, R)$.

So $a_n - \overline{a_n} = 0$ for all n .

#8.2.2. Let $p(z) = a_n z^n + \dots + a_0$ ($a_n \neq 0$).

Given $\varepsilon > 0$, there exists $R > 0$ such that

$$r > R \Rightarrow (1 - \varepsilon) \cdot |a_n| \cdot r^n \leq |p(z)| \leq (1 + \varepsilon) |a_n| \cdot r^n \quad \text{for all } |z| = r.$$

#8.2.3.

If all the zeros of a polynomial $P(z)$ have positive real part, then all the zeros of $P'(z)$ have positive real part.

Proof. Denote that $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 = a_n (z - z_1) \dots (z - z_n)$.

Then, $P'(z) = a_n \cdot \left[\sum_{i=1}^n \prod_{j \neq i} (z - z_j) \right]$, so therefore

$$\frac{P'(z)}{P(z)} = \sum_{i=1}^n \frac{1}{z - z_i} \quad \text{for } z \neq z_1, \dots, z_n.$$

If $\operatorname{Re} z \leq 0$, then $\operatorname{Re}(z - z_i) < 0$ for all $1 \leq i \leq n$, so $P'(z)/P(z)$ cannot be zero, so $P'(z)$ cannot be zero. Therefore, if $P'(z) = 0$, $\operatorname{Re} z > 0$.

#8.2.4.

Let f be an entire satisfying $|f(z)| \leq |e^z|$ for all $z \in \mathbb{C}$.

Show that $f(z) = k e^z$, $|k| \leq 1$.

Proof. Let $g(z) = f(z)/e^z$. Since f is entire, so is g .

And, $|g(z)| = |f(z)|/|e^z| \leq 1$, so $g(z)$ must be a constant, by the Liouville's Thm.

Now, $f(z) = g(z) \cdot e^z$, where g is a constant satisfying $|g| \leq 1$.

#8.2.6.

Let f be an entire s.t. $|f'(z)| \leq |z|$ for all $z \in \mathbb{C}$.

Then $f(z) = az^2 + b$, where $a, b \in \mathbb{C}$ with $|a| \leq \frac{1}{2}$.

Proof. Given $z_0 \in \mathbb{C}$, and $R > 0$, $|f'(z)| \leq |z| = |z - z_0 + z_0| \leq R + |z_0|$, for $|z - z_0| = R$.

So, $|f''(z_0)| \leq \frac{1!}{R} \cdot \sup_{|z - z_0| = R} |f'(z)| \leq \frac{R + |z_0|}{R} = 1 + \frac{|z_0|}{R} \rightarrow 1$ as $R \rightarrow \infty$.

and, for $n \geq 3$,

$$|f^{(n)}(z_0)| \leq \frac{(n-1)!}{R^{n-1}} \cdot \sup |f'| \leq \frac{R + |z_0|}{R^{n-1}} \rightarrow 0 \text{ as } R \rightarrow \infty.$$

$$\text{And, } |f'(z_0)| \leq \frac{1}{2\pi} \int_{\mathbb{C}} \frac{|f(z)|}{z} dz \leq \frac{2\pi R}{2\pi} \rightarrow 0 \text{ as } R \rightarrow \infty,$$

Sec 9.2.

#1. Suppose that $f(z)$ has an iso. sing. at z_0 .

If $\lim_{z \rightarrow z_0} (z - z_0)^\alpha \cdot f(z) = M$ or ∞ , where $M \neq 0$, then α must be integer.

proof Suppose that $\lim_{z \rightarrow z_0} (z - z_0)^\alpha \cdot f(z) = M$. Then, given $\epsilon > 0$, there is $\delta > 0$ s.t.
 $0 < |z - z_0| < \delta \Rightarrow |(z - z_0)^\alpha \cdot f(z)| < |M| + \epsilon$.

So, the $(z - z_0)^\alpha \cdot f(z)$ is bounded in the punctured $0 < |z - z_0| < \delta$

Then, $(z - z_0)^\alpha \cdot f(z)$ has a Taylor Expression

$$(z - z_0)^\alpha f(z) = a_0 + a_1(z - z_0) + \frac{1}{2!} \cdot a_2(z - z_0)^2 + \dots$$

#2. If f is analytic in $B_R(0) \setminus \{0\}$ and $\lim_{z \rightarrow 0} |zf(z)| = 0$.

Then the origin is removable.

proof. Since

